CAS SciFinder

Task History

Initiating Search

November 10, 2025, 4:21 PM

Search:

CAS Lexicon Search:

Transhalogenation - Preferred Concept

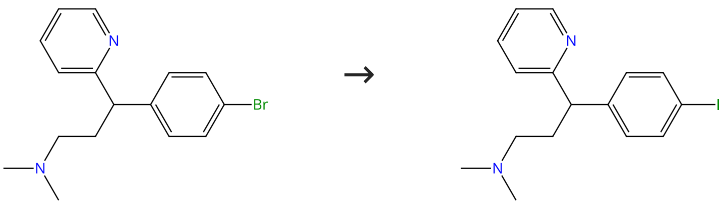
Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (579)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (250) Filtered By: Yield: **90-100%, 80-89%, 70-79%** Document Type: **Journal**	**Reactions**	View Results

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (34)

31-174-CAS-14443138 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium iodide
Catalysts: Cuprous iodide, *trans*-*N*,*N'*-Dimethyl-1,2-cyclohexanediamine
Solvents: 1,4-Dioxane; 24 h, 110 °C; 110 °C → rt

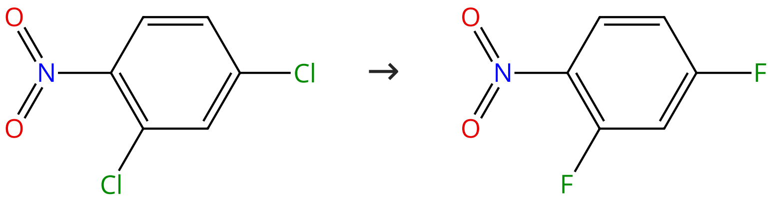
Experimental Protocols

Copper-Catalyzed Halogen Exchange in Aryl Halides: An Aromatic Finkelstein Reaction

By: Klapars, Artis; et al

Journal of the American Chemical Society (2002), 124(50), 14844-14845.

Scheme 2 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (62) Suppliers (86)

31-174-CAS-14529133 Steps: **1** Yield: **100%**

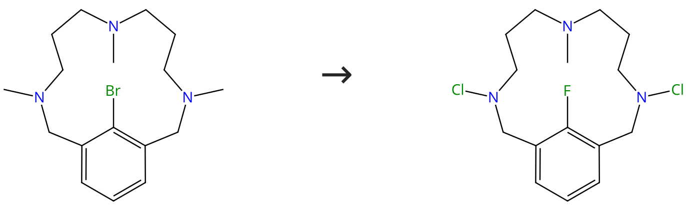
1.1 **Reagents:** Potassium fluoride
Catalysts: Trihexyl(tetradecyl)phosphonium tetrafluoroborate
Solvents: Dimethyl sulfoxide; 1.8 h, 180 °C

Phosphonium ionic liquids as highly thermal stable and efficient phase transfer catalysts for solid-liquid Halex reactions

By: Fan, Ao; et al

Catalysis Today (2012), 198(1), 300-304.

Scheme 3 (4 Reactions) Steps: **1** Yield: **100%**



31-174-CAS-5970042 Steps: **1** Yield: **100%**

1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Silver fluoride
Catalysts: Chloro(triphenylphosphine)gold
Solvents: Acetonitrile-d_3; 48 h, 40 °C

Experimental Protocols

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

31-174-CAS-12366314	Steps: **1** Yield: **100%**
1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Silver fluoride **Solvents:** Acetonitrile-d_3; 48 h, 40 °C	
Experimental Protocols	

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

31-174-CAS-10236393	Steps: **1** Yield: **100%**
1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Silver fluoride **Catalysts:** Gold(1+), (acetonitrile)[1,3-bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]-, (*OC*-6-11)-hexafluoroantimonate(1-) (1:1) **Solvents:** Acetonitrile-d_3; 6 h, 40 °C	
Experimental Protocols	

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

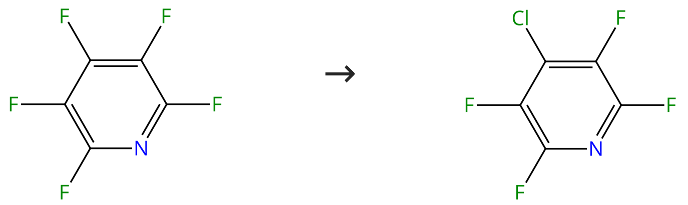
31-174-CAS-8102564	Steps: **1** Yield: **100%**
1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Silver fluoride **Catalysts:** [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl]methanesulfonamidato-κ*N*](triphenylphosphine)gold **Solvents:** Acetonitrile-d_3; 48 h, 40 °C	
Experimental Protocols	

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (71) Suppliers (32)

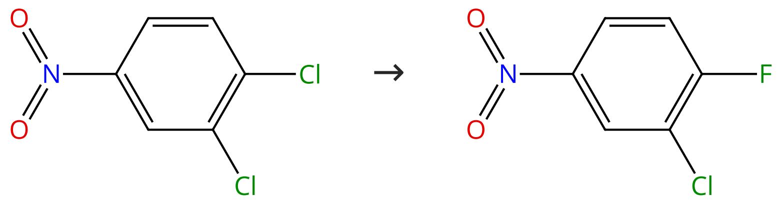
31-174-CAS-16538737	Steps: **1** Yield: **100%**
1.1 **Reagents:** Lithium chloride **Solvents:** Dimethyl sulfoxide; 48 h, 80 °C	
Experimental Protocols	

Computational and Experimental Studies of Regioselective S_NAr Halide Exchange (Halex) Reactions of Pentachloropyridine

By: Froese, Robert D. J.; et al

Journal of Organic Chemistry (2016), 81(22), 10672-10682.

Scheme 5 (2 Reactions) Steps: **1** Yield: **97-100%**



Suppliers (61) Suppliers (85)

31-174-CAS-11906218	Steps: **1** Yield: **100%**
1.1 **Reagents:** Potassium fluoride **Catalysts:** Bis(triphenylphosphine)iminium chloride **Solvents:** Dimethyl sulfoxide; 8 h, 150 °C; cooled	
Experimental Protocols	

Efficient phosphorus catalysts for the halogen-exchange (Halex) reaction

By: Lacour, Marie-Agnes; et al

Advanced Synthesis & Catalysis (2008), 350(17), 2677-2682.

31-174-CAS-7687920	Steps: **1** Yield: **97%**

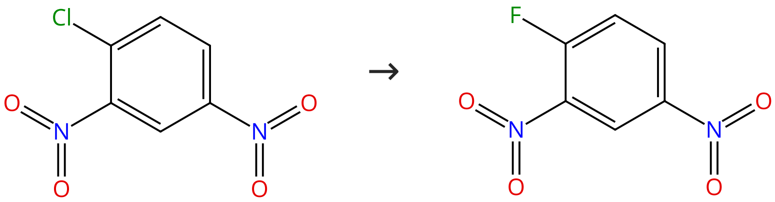
1.1 **Reagents:** Potassium fluoride
Catalysts: Antimony trichloride
Solvents: Dimethyl sulfoxide; 10 min, heated

Synthesis of 3-chloro-4-fluoronitrobenzene promoted by microwave irradiation

By: Luo, Jun; et al

Jingxi Huagong Zhongjianti (2007), 37(1), 30-33.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (75) Suppliers (75)

31-174-CAS-15457454	Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium fluoride
Catalysts: Bis(triphenylphosphine)iminium chloride
Solvents: Dimethyl sulfoxide; 8 h, 150 °C; cooled

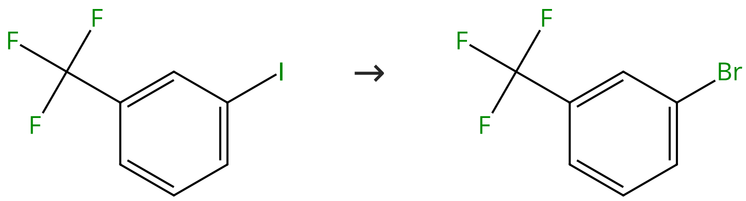
Experimental Protocols

Efficient phosphorus catalysts for the halogen-exchange (Halex) reaction

By: Lacour, Marie-Agnes; et al

Advanced Synthesis & Catalysis (2008), 350(17), 2677-2682.

Scheme 7 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (74) Suppliers (85)

31-614-CAS-33927301	Steps: **1** Yield: **99%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: 1,4-Dioxane; 10 min, rt

1.2 **Reagents:** Tetrabutylammonium bromide
Solvents: 1,4-Dioxane; 36 h, rt

1.3 **Reagents:** Water; rt

Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 8 (1 Reaction) Steps: **1** Yield: **99%**

Suppliers (87) Suppliers (66)

31-614-CAS-33927278 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Cyclopentyl methyl ether; 10 min, rt

1.2 **Reagents:** Sodium iodide
Solvents: Cyclopentyl methyl ether; 36 h, 50 °C

1.3 **Reagents:** Water; rt

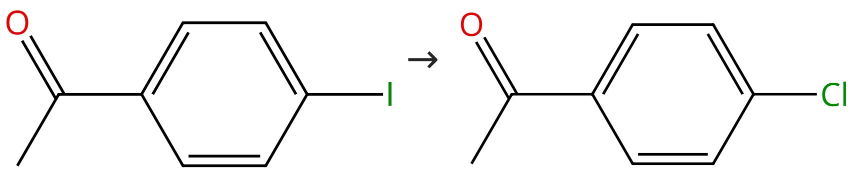
Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 9 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (87) Suppliers (67)

31-614-CAS-33927315 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: 1,4-Dioxane; 10 min, rt

1.2 **Reagents:** Benzyltriethylammonium chloride
Solvents: 1,4-Dioxane; 24 h, rt

1.3 **Reagents:** Water; rt

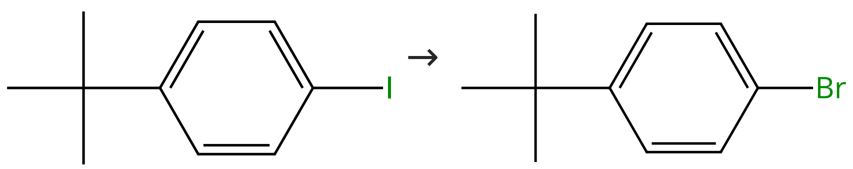
Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 10 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (68) Suppliers (78)

31-614-CAS-33927294 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: 1,4-Dioxane; 10 min, rt

1.2 **Reagents:** Tetrabutylammonium bromide
Solvents: 1,4-Dioxane; 36 h, rt

1.3 **Reagents:** Water; rt

Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 11 (1 Reaction) Steps: **1** Yield: **99%**

Suppliers (76) Suppliers (75)

31-614-CAS-33927287 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Cyclopentyl methyl ether; 10 min, rt

1.2 **Reagents:** Sodium iodide
Solvents: Cyclopentyl methyl ether; 36 h, 50 °C

1.3 **Reagents:** Water; rt

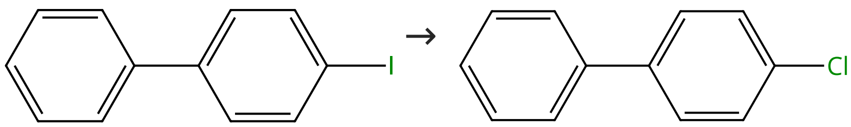
Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 12 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (85) Suppliers (59)

31-614-CAS-33927307 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: 1,4-Dioxane; 10 min, rt

1.2 **Reagents:** Benzyltriethylammonium chloride
Solvents: 1,4-Dioxane; 36 h, rt

1.3 **Reagents:** Water; rt

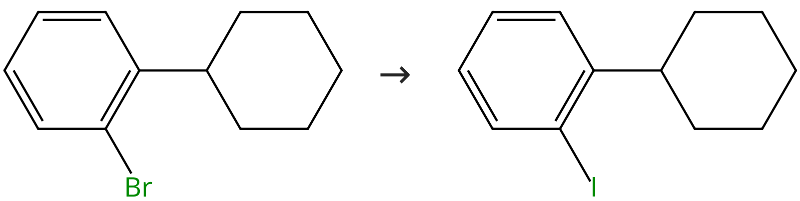
Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 13 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (50) Suppliers (11)

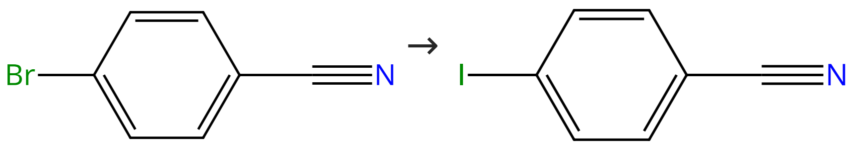
31-174-CAS-3475487	Steps: **1** Yield: **99%**
1.1	**Reagents:** Sodium iodide **Catalysts:** Cuprous iodide, *trans*-*N*,*N*'-Dimethyl-1,2-cyclohexanediamine **Solvents:** 1-Pentanol; 40 h, 130 °C; 130 °C → rt
1.2	**Solvents:** Hexane
Experimental Protocols	

Copper-Catalyzed Halogen Exchange in Aryl Halides: An Aromatic Finkelstein Reaction

By: Klapars, Artis; et al

Journal of the American Chemical Society (2002), 124(50), 14844-14845.

Scheme 14 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (92) Suppliers (80)

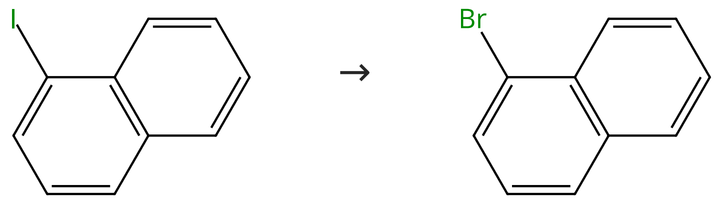
31-614-CAS-33927274	Steps: **1** Yield: **99%**
1.1	**Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine **Solvents:** Cyclopentyl methyl ether; 10 min, rt
1.2	**Reagents:** Sodium iodide **Solvents:** Cyclopentyl methyl ether; 36 h, 50 °C
1.3	**Reagents:** Water; rt
Experimental Protocols	

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 15 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (85) Suppliers (91)

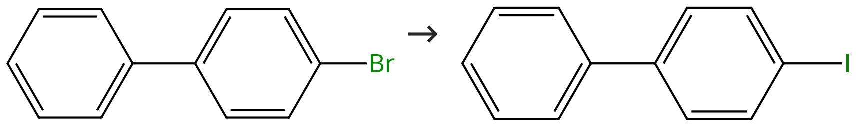
31-614-CAS-33927311	Steps: **1** Yield: **99%**
1.1	**Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine **Solvents:** 1,4-Dioxane; 10 min, rt
1.2	**Reagents:** Tetrabutylammonium bromide **Solvents:** 1,4-Dioxane; 36 h, 50 °C
1.3	**Reagents:** Water; rt
Experimental Protocols	

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 16 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (89)

Suppliers (85)

31-614-CAS-33927271 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Cyclopentyl methyl ether; 10 min, rt

1.2 **Reagents:** Sodium iodide
Solvents: Cyclopentyl methyl ether; 12 h, 50 °C

1.3 **Reagents:** Water; rt

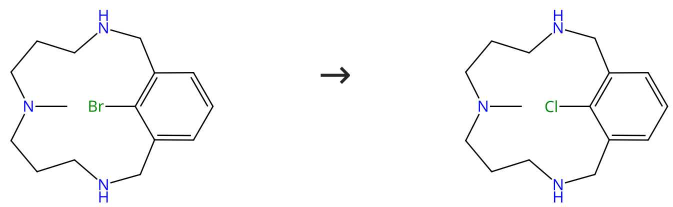
Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 17 (3 Reactions) Steps: **1** Yield: **99%**



31-174-CAS-6303381 Steps: **1** Yield: **99%**

1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Tetrabutylammonium chloride
Catalysts: [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl]methanesulfonamidato-κ*N*](triphenylphosphine)gold
Solvents: Acetonitrile-d_3; 60 h, 40 °C

Experimental Protocols

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

31-174-CAS-8427690 Steps: **1** Yield: **99%**

1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Tetrabutylammonium chloride
Catalysts: Gold(1+), (acetonitrile)[1,3-bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]-, (*OC*-6-11)-hexafluoroantimonate(1-) (1:1)
Solvents: Acetonitrile-d_3; 12 h, 40 °C

Experimental Protocols

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

31-174-CAS-2651901 Steps: **1** Yield: **99%**

1.1 **Reagents:** Tetrabutylammonium chloride
Catalysts: Copper(1+), tetrakis(acetonitrile)-, (*T*-4)-, 1,1,1-trifluoromethanesulfonate (1:1)
Solvents: Acetonitrile-d_3; 2 h, 25 °C

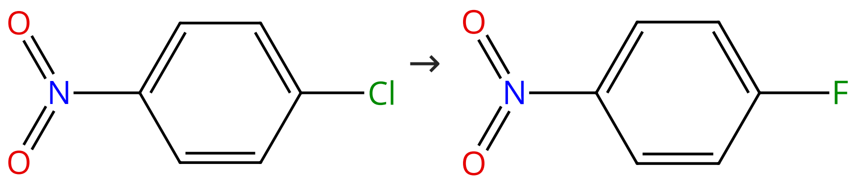
Experimental Protocols

Nucleophilic Aryl Fluorination and Aryl Halide Exchange Mediated by a Cu^{I}/Cu^{III} Catalytic Cycle

By: Casitas, Alicia; et al

Journal of the American Chemical Society (2011), 133(48), 19386-19392.

Scheme 18 (2 Reactions) Steps: **1** Yield: **97-99%**



Suppliers (65) Suppliers (99)

31-174-CAS-643431 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium fluoride
Catalysts: Bis(triphenylphosphine)iminium chloride
Solvents: Dimethyl sulfoxide; 8 h, 150 °C; cooled

Experimental Protocols

Efficient phosphorus catalysts for the halogen-exchange (Halex) reaction

By: Lacour, Marie-Agnes; et al

Advanced Synthesis & Catalysis (2008), 350(17), 2677-2682.

31-174-CAS-10555300 Steps: **1** Yield: **97%**

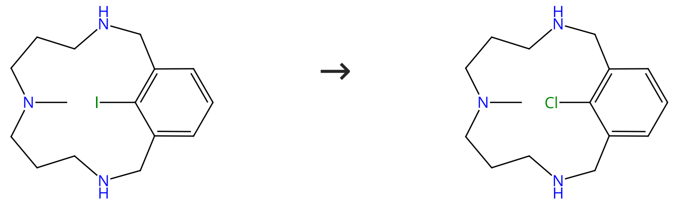
1.1 **Reagents:** Potassium fluoride
Catalysts: Trihexyl(tetradecyl)phosphonium tetrafluoroborate
Solvents: Dimethyl sulfoxide; 9 h, 180 °C

Phosphonium ionic liquids as highly thermal stable and efficient phase transfer catalysts for solid-liquid Halex reactions

By: Fan, Ao; et al

Catalysis Today (2012), 198(1), 300-304.

Scheme 19 (3 Reactions) Steps: **1** Yield: **99%**



31-174-CAS-11859178 Steps: **1** Yield: **99%**

1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Tetrabutylammonium chloride
Catalysts: Gold(1+), (acetonitrile)[1,3-bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]-, (*OC*-6-11)-hexafluoroantimonate(1-) (1:1)
Solvents: Acetonitrile-d_3; 6 h, 40 °C

Experimental Protocols

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

31-174-CAS-9731251 Steps: **1** Yield: **99%**

1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Tetrabutylammonium chloride
Catalysts: Chloro(triphenylphosphine)gold
Solvents: Acetonitrile-d_3; 48 h, 40 °C

Experimental Protocols

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

31-174-CAS-11772369 Steps: **1** Yield: **99%**

1.1 **Reagents:** Tetrabutylammonium chloride
Catalysts: Copper(1+), tetrakis(acetonitrile)-, (*T*-4)-, 1,1,1-trifluoromethanesulfonate (1:1)
Solvents: Acetonitrile-d_3; 1.5 h, 25 °C

Experimental Protocols

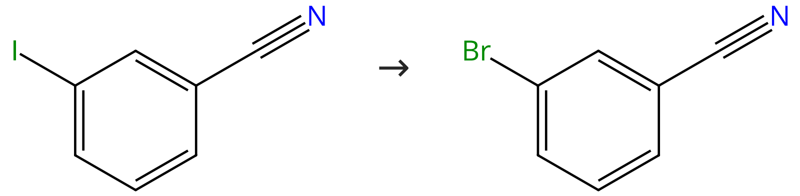
Nucleophilic Aryl Fluorination and Aryl Halide Exchange Mediated by a Cu^{I}/Cu^{III} Catalytic Cycle

By: Casitas, Alicia; et al

Journal of the American Chemical Society (2011), 133(48), 19386-19392.

Scheme 20 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (85)

Suppliers (79)

31-614-CAS-33927299

Steps: **1** Yield: **99%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: 1,4-Dioxane; 10 min, rt

1.2 **Reagents:** Tetrabutylammonium bromide
Solvents: 1,4-Dioxane; 36 h, rt

1.3 **Reagents:** Water; rt

Experimental Protocols

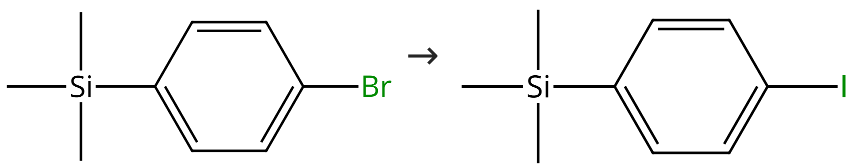
Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 21 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (67)

Suppliers (14)

31-614-CAS-33927273

Steps: **1** Yield: **99%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Cyclopentyl methyl ether; 10 min, rt

1.2 **Reagents:** Sodium iodide
Solvents: Cyclopentyl methyl ether; 36 h, 50 °C

1.3 **Reagents:** Water; rt

Experimental Protocols

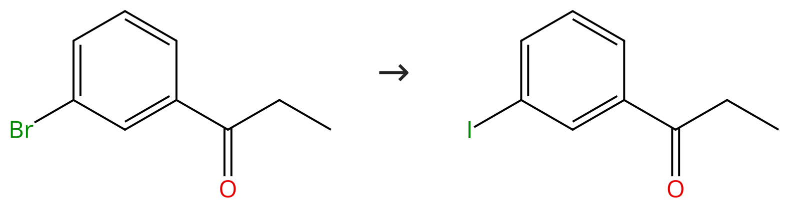
Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 22 (1 Reaction)

Steps: **1** Yield: **98%**



Suppliers (80)

Suppliers (19)

31-174-CAS-4451592 Steps: **1** Yield: **98%**

1.1 **Reagents:** Sodium iodide
Catalysts: Cuprous iodide, *trans*-*N*,*N*'-Dimethyl-1,2-cyclohexanediamine
Solvents: 1,4-Dioxane; 24 h, 110 °C; 110 °C → rt

1.2 **Reagents:** Ammonia
Solvents: Water; 15 min, rt; 15 h, rt

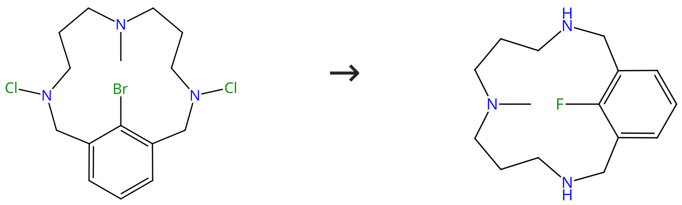
Experimental Protocols

Copper-Catalyzed Halogen Exchange in Aryl Halides: An Aromatic Finkelstein Reaction

By: Klapars, Artis; et al

Journal of the American Chemical Society (2002), 124(50), 14844-14845.

Scheme 23 (1 Reaction) Steps: **1** Yield: **98%**



31-174-CAS-7825553 Steps: **1** Yield: **98%**

1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Silver fluoride
Catalysts: Gold(1+), (acetonitrile)[1,3-bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]-, (*OC*-6-11)-hexafluoroantimonate(1-) (1:1)
Solvents: Acetonitrile-d_3; 12 h, 70 °C

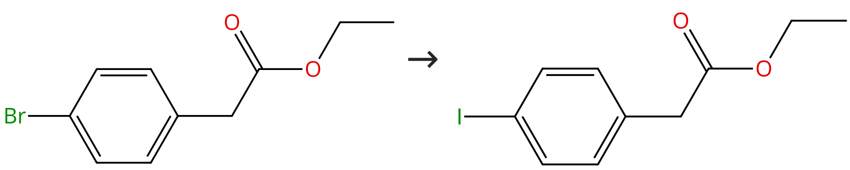
Experimental Protocols

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

Scheme 24 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (88) Suppliers (60)

31-174-CAS-12972100 Steps: **1** Yield: **98%**

1.1 **Reagents:** Sodium iodide
Catalysts: Cuprous iodide, *trans*-*N*,*N*'-Dimethyl-1,2-cyclohexanediamine
Solvents: 1,4-Dioxane; 24 h, 110 °C; 110 °C → rt

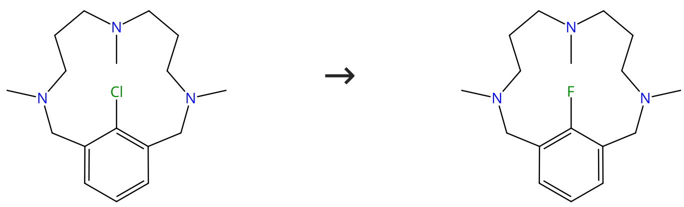
Experimental Protocols

Copper-Catalyzed Halogen Exchange in Aryl Halides: An Aromatic Finkelstein Reaction

By: Klapars, Artis; et al

Journal of the American Chemical Society (2002), 124(50), 14844-14845.

Scheme 25 (1 Reaction) Steps: **1** Yield: **98%**



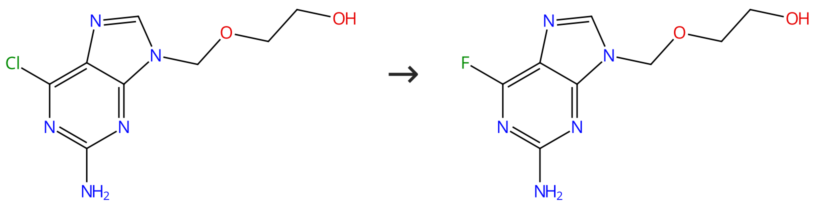
31-174-CAS-12004546	Steps: **1** Yield: **98%**
1.1	**Reagents:** Silver fluoride **Catalysts:** Copper(1+), tetrakis(acetonitrile)-, (*T*-4)-, 1,1,1-trifluoromethanesulfonate (1:1) **Solvents:** Acetone, Acetonitrile; 12 h, 298 K
Experimental Protocols	

Nucleophilic Aryl Fluorination and Aryl Halide Exchange Mediated by a Cu^{I}/Cu^{III} Catalytic Cycle

By: Casitas, Alicia; et al

Journal of the American Chemical Society (2011), 133(48), 19386-19392.

Scheme 26 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (5) Suppliers (7)

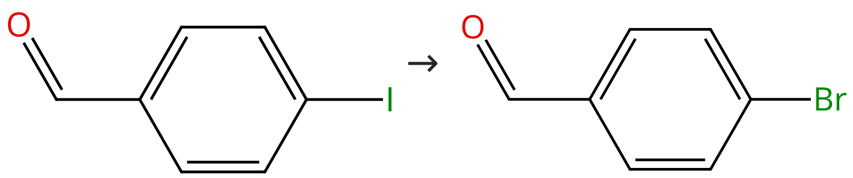
31-174-CAS-15357477	Steps: **1** Yield: **98%**
1.1	**Reagents:** Potassium fluoride **Catalysts:** Triethylamine **Solvents:** 1-Butyl-3-methylimidazolium tetrafluoroborate; 10 min, 100 °C

Green synthesis of 6-fluoropurine under microwave irradi ation

By: Xu, Shao-hong; et al

Huaxue Shiji (2010), 32(9), 862-864.

Scheme 27 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (92) Suppliers (91)

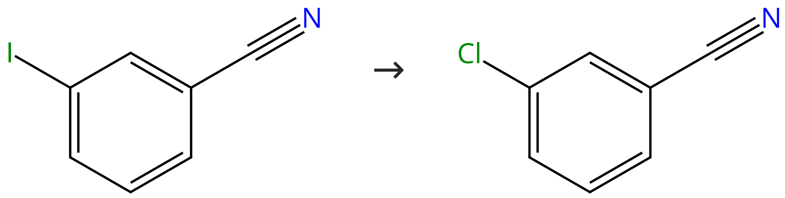
31-614-CAS-33927291	Steps: **1** Yield: **98%**
1.1	**Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethy lethyl)-2,2'-bipyridine **Solvents:** 1,4-Dioxane; 10 min, rt
1.2	**Reagents:** Tetrabutylammonium bromide **Solvents:** 1,4-Dioxane; 36 h, rt
1.3	**Reagents:** Water; rt
Experimental Protocols	

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

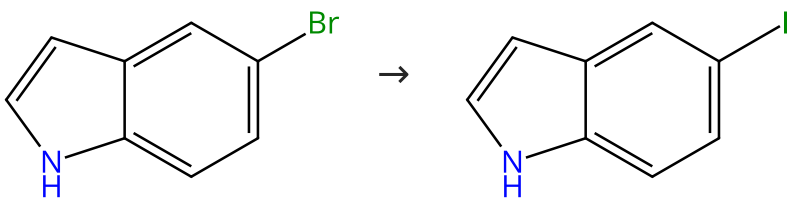
Scheme 28 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (85) Suppliers (73)

31-614-CAS-33927317	Steps: **1** Yield: **98%**	Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange
1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine **Solvents:** 1,4-Dioxane; 10 min, rt 1.2 **Reagents:** Benzyltriethylammonium chloride **Solvents:** 1,4-Dioxane; 36 h, 60 °C 1.3 **Reagents:** Water; rt		By: Feng, Yunhui; et al ACS Catalysis (2022), 12(18), 11089-11096.
Experimental Protocols		

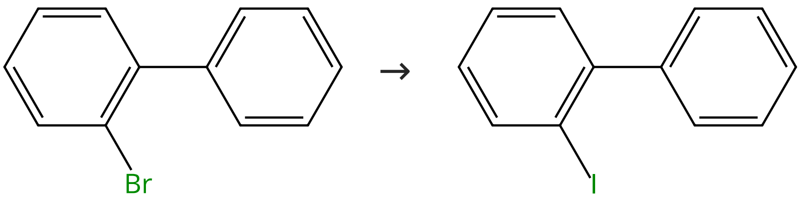
Scheme 29 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (108) Suppliers (72)

31-174-CAS-3525315	Steps: **1** Yield: **98%**	Copper-Catalyzed Halogen Exchange in Aryl Halides: An Aromatic Finkelstein Reaction
1.1 **Reagents:** Sodium iodide **Catalysts:** Cuprous iodide, *trans*-*N*,*N'*-Dimethyl-1,2-cyclohexanediamine **Solvents:** 1,4-Dioxane; 24 h, 110 °C; 110 °C → rt		By: Klapars, Artis; et al Journal of the American Chemical Society (2002), 124(50), 14844-14845.
Experimental Protocols		

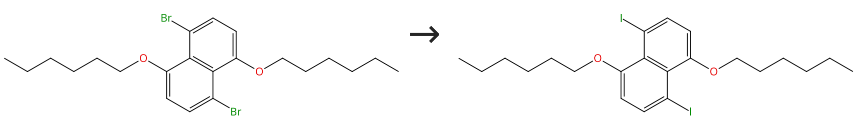
Scheme 30 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (84) Suppliers (79)

31-614-CAS-33927283	Steps: **1** Yield: **98%**	Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange
1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine **Solvents:** Cyclopentyl methyl ether; 10 min, rt 1.2 **Reagents:** Sodium iodide **Solvents:** Cyclopentyl methyl ether; 36 h, 50 °C 1.3 **Reagents:** Water; rt		By: Feng, Yunhui; et al ACS Catalysis (2022), 12(18), 11089-11096.
Experimental Protocols		

Scheme 31 (1 Reaction) Steps: **1** Yield: **97%**



Supplier (1) Supplier (1)

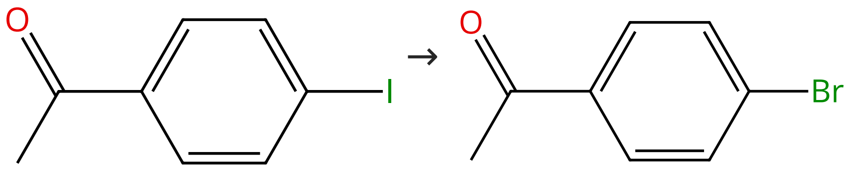
31-174-CAS-2379465	Steps: **1** Yield: **97%**
1.1	**Reagents:** Butyllithium **Solvents:** Tetrahydrofuran, Hexane
1.2	**Reagents:** Iodine **Solvents:** Tetrahydrofuran

A convenient synthetic approach to 1,5-diiodonaphthalene derivatives

By: Pan, Yongchun; et al

Tetrahedron Letters (2000), 41(23), 4537-4540.

Scheme 32 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (87) Suppliers (84)

31-614-CAS-33927296	Steps: **1** Yield: **97%**
1.1	**Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine **Solvents:** 1,4-Dioxane; 10 min, rt
1.2	**Reagents:** Tetrabutylammonium bromide **Solvents:** 1,4-Dioxane; 12 h, rt
1.3	**Reagents:** Water; rt

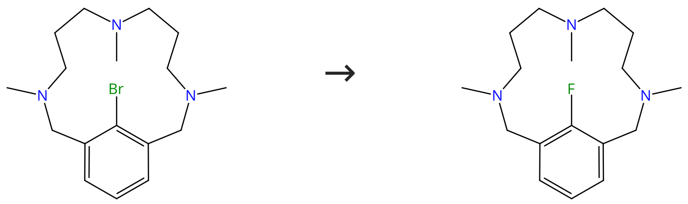
Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 33 (1 Reaction) Steps: **1** Yield: **97%**



31-174-CAS-14134337	Steps: **1** Yield: **97%**
1.1	**Reagents:** Silver fluoride **Catalysts:** Copper(1+), tetrakis(acetonitrile)-, (*T*-4)-, 1,1,1-trifluoromethanesulfonate (1:1) **Solvents:** Acetone, Acetonitrile; 24 h, 298 K

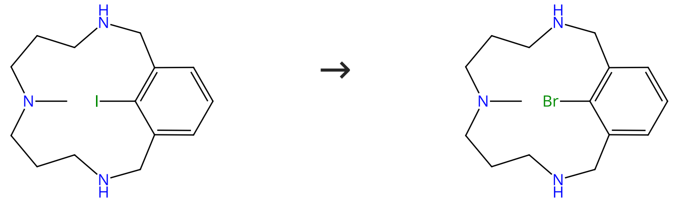
Experimental Protocols

Nucleophilic Aryl Fluorination and Aryl Halide Exchange Mediated by a Cu^{I}/Cu^{III} Catalytic Cycle

By: Casitas, Alicia; et al

Journal of the American Chemical Society (2011), 133(48), 19386-19392.

Scheme 34 (1 Reaction) Steps: **1** Yield: **97%**



31-174-CAS-13981989 Steps: **1** Yield: **97%**

1.1 **Reagents:** 1,3,5-Trimethoxybenzene, Sodium bromide
Catalysts: Gold(1+), (acetonitrile)[1,3-bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]-, (*OC*-6-11)-hexafluoroantimonate(1-) (1:1)
Solvents: Acetonitrile-d_3; 30 h, 70 °C

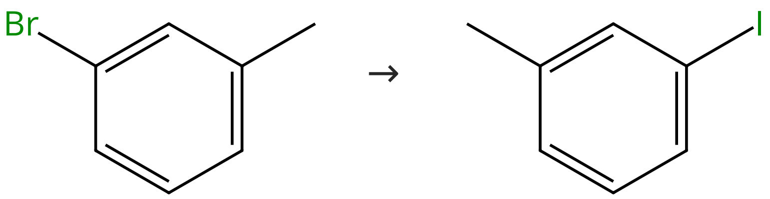
Experimental Protocols

Oxidant-Free Au(I)-Catalyzed Halide Exchange and C_{sp2}-O Bond Forming Reactions

By: Serra, Jordi; et al

Journal of the American Chemical Society (2015), 137(41), 13389-13397.

Scheme 35 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (58) Suppliers (73)

31-614-CAS-33927281 Steps: **1** Yield: **97%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Cyclopentyl methyl ether; 10 min, rt

1.2 **Reagents:** Sodium iodide
Solvents: Cyclopentyl methyl ether; 36 h, 50 °C

1.3 **Reagents:** Water; rt

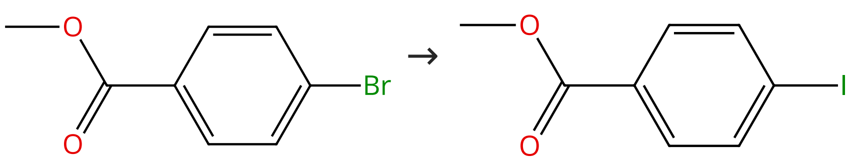
Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 36 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (97) Suppliers (91)

31-614-CAS-33927267 Steps: **1** Yield: **97%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Cyclopentyl methyl ether; 10 min, rt

1.2 **Reagents:** Sodium iodide
Solvents: Cyclopentyl methyl ether; 36 h, 50 °C

1.3 **Reagents:** Water; rt

Experimental Protocols

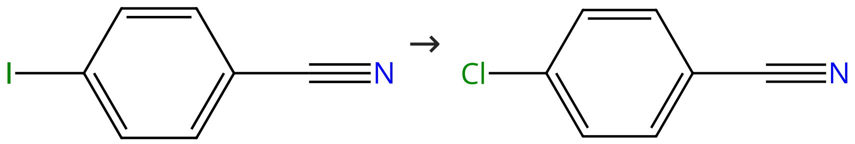
Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 37 (1 Reaction)

Steps: **1** Yield: **97%**



Suppliers (80)

Suppliers (89)

31-614-CAS-33927309

Steps: **1** Yield: **97%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: 1,4-Dioxane; 10 min, rt

1.2 **Reagents:** Benzyltriethylammonium chloride
Solvents: 1,4-Dioxane; 48 h, rt

1.3 **Reagents:** Water; rt

Experimental Protocols

Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 38 (1 Reaction)

Steps: **1** Yield: **97%**

Suppliers (74)

Suppliers (86)

31-614-CAS-33927312

Steps: **1** Yield: **97%**

1.1 **Catalysts:** Bis(1,5-cyclooctadiene)nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: 1,4-Dioxane; 10 min, rt

1.2 **Reagents:** Tetrabutylammonium bromide
Solvents: 1,4-Dioxane; 48 h, 50 °C

1.3 **Reagents:** Water; rt

Experimental Protocols

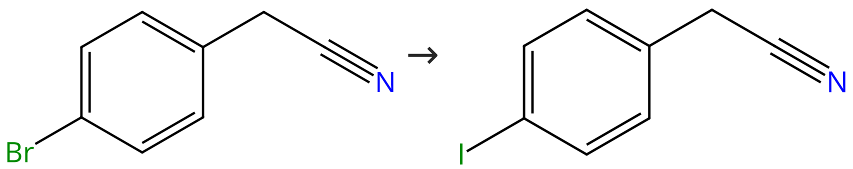
Light-Promoted Nickel-Catalyzed Aromatic Halogen Exchange

By: Feng, Yunhui; et al

ACS Catalysis (2022), 12(18), 11089-11096.

Scheme 39 (1 Reaction)

Steps: **1** Yield: **97%**



Suppliers (70)

Suppliers (68)

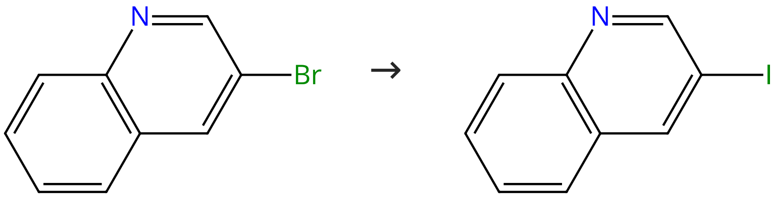
31-174-CAS-8715362	Steps: **1** Yield: **97%**
1.1	**Reagents:** Sodium iodide **Catalysts:** Cuprous iodide, *trans-N,N'*-Dimethyl-1,2-cyclohexanediamine **Solvents:** 1,4-Dioxane; 24 h, 110 °C; 110 °C → rt
1.2	**Reagents:** Ammonia **Solvents:** Water; 15 min, rt; 15 h, rt
Experimental Protocols	

Copper-Catalyzed Halogen Exchange in Aryl Halides: An Aromatic Finkelstein Reaction

By: Klapars, Artis; et al

Journal of the American Chemical Society (2002), 124(50), 14844-14845.

Scheme 40 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (97) Suppliers (57)

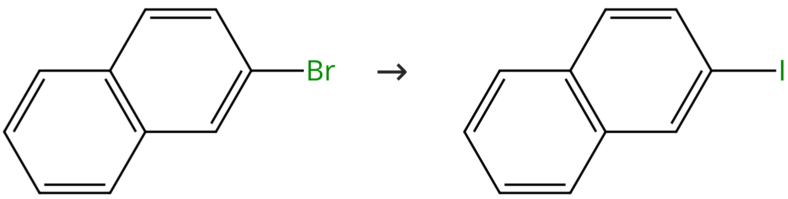
31-174-CAS-8061562	Steps: **1** Yield: **97%**
1.1	**Reagents:** Sodium iodide **Catalysts:** Cuprous iodide, *trans-N,N'*-Dimethyl-1,2-cyclohexanediamine **Solvents:** 1,4-Dioxane; 24 h, 110 °C; 110 °C → rt
Experimental Protocols	

Copper-Catalyzed Halogen Exchange in Aryl Halides: An Aromatic Finkelstein Reaction

By: Klapars, Artis; et al

Journal of the American Chemical Society (2002), 124(50), 14844-14845.

Scheme 41 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (87) Suppliers (65)

31-174-CAS-7777506	Steps: **1** Yield: **97%**
1.1	**Reagents:** Sodium iodide **Catalysts:** Cuprous iodide, *trans-N,N'*-Dimethyl-1,2-cyclohexanediamine **Solvents:** Acetonitrile; 2 min, rt → 100 °C; 60 min, 100 °C
Experimental Protocols	

Hydrodehalogenation of Aryl Chlorides and Aryl Bromides Using a Microwave-Assisted, Copper-Catalyzed Concurrent Tandem Catalysis Methodology

By: Cannon, Kathleen A.; et al

Organometallics (2011), 30(15), 4067-4073.